NOETHERIANNESS OF LOCAL RINGS IS PRESERVED IN BOTH DIRECTIONS ALONG AN ETALE MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Noetherianness descends along a faithfully flat ring map. the author cites Algebra, Lemma algebra-lemma-descent-Noetherian at proof lines 10823-10825. Gives the direction from the target to the source.
- An etale localised map is faithfully flat, and a localisation of a finite type algebra over a Noetherian ring is Noetherian. used at proof lines 10820-10822 and 10826-10827. Provides the flatness used in the first direction and the finite type argument in the second.

Source reference. Original lemma 0AGZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13212–13212. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 033E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4053–4053. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *If $A \to B$ is an étale ring map and $\mathfrak{q}$ is a prime of B lying over $\mathfrak{p} \subset A$, then $A_\mathfrak{p}$ is Noetherian if and only if $B_\mathfrak{q}$ is Noetherian.*

Proof. Since $A_\mathfrak{p} \to B_\mathfrak{q}$ is faithfully flat we see that $B_\mathfrak{q}$ Noetherian implies that $A_\mathfrak{p}$ is Noetherian, see Algebra, Lemma 033E. Conversely, if $A_\mathfrak{p}$ is Noetherian, then $B_\mathfrak{q}$ is Noetherian as it is a localization of a finite type $A_\mathfrak{p}$-algebra. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.